

The Periodic Table: chemical periodicity

A Level Chemistry

Physical properties across Period 3

Periodicity 周期性 means that properties repeat in a regular pattern as you go across each **period** 周期 of the Periodic Table. Period 3 (Na to Ar) is the standard example.

Property	Trend across Period 3
atomic radius 原子半径	gets smaller (more nuclear charge pulls the same shell in)
ionic radius 离子半径	positive ions are small; from P^{3-} onwards the negative ions are larger
melting point 熔点	rises to a peak at silicon, then falls sharply
electrical conductivity 导电性	high for Na, Mg, Al; almost zero from Si onwards

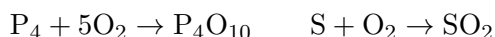
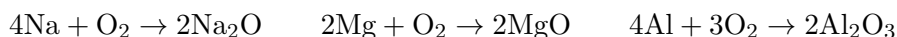
The melting point and conductivity follow from the structure and bonding:

- **Na, Mg, Al** are **giant metallic** 金属晶体. Melting points rise (Na $\rightarrow$ Al) because each atom gives more delocalised electrons and the ions get smaller, so the bonding is stronger. They conduct well.
- **Si** is **giant molecular** 原子晶体 (giant covalent). It has the highest melting point, because strong covalent bonds must be broken. It barely conducts.
- **P, S, Cl, Ar** are **simple molecular** 分子晶体 (or single atoms). Their melting points are low, because only weak intermolecular forces break. They do not conduct.

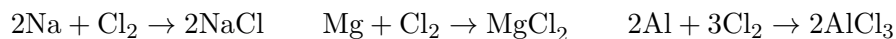
Chemical properties across Period 3

Reactions with oxygen, chlorine and water

With **oxygen**:



With **chlorine**:



With **water** (only Na and Mg react):



Sodium reacts fast; magnesium reacts only very slowly with cold water.

Oxidation number of the oxides and chlorides

The **oxidation number** 氧化数 of the Period 3 element in its oxide or chloride rises across the period, because it equals the number of outer-shell (**valence shell** 价层) **electrons** 电子 the atom uses in bonding:

Oxides	Na ₂ O	MgO	Al ₂ O ₃	P ₄ O ₁₀	SO ₂ / SO ₃
oxidation number	+1	+2	+3	+5	+4 / +6

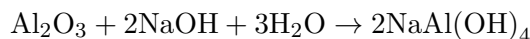
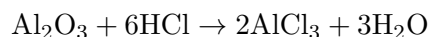
The chlorides NaCl, MgCl₂, AlCl₃, SiCl₄, PCl₅ show oxidation numbers +1 to +5 in the same way.

Oxides with water, and acid–base behaviour

Across the period the **oxides** 氧化物 change from basic to acidic:

Oxide	With water	Acid–base nature	Approximate pH
Na ₂ O	Na ₂ O + H ₂ O → 2NaOH	basic	13–14
MgO	MgO + H ₂ O → Mg(OH) ₂	basic	9–10
Al ₂ O ₃	insoluble	amphoteric 两性	7
SiO ₂	insoluble	weakly acidic	7
P ₄ O ₁₀	P ₄ O ₁₀ + 6H ₂ O → 4H ₃ PO ₄	acidic	1–2
SO ₃	SO ₃ + H ₂ O → H ₂ SO ₄	strongly acidic	0–1

Metal oxides (left) are basic; non-metal oxides (right) are acidic. Al₂O₃ and its **hydroxide** 氢氧化物 Al(OH)₃ are amphoteric—they react with **both** acids and bases:



Chlorides with water

- NaCl and MgCl₂ are ionic. They simply dissolve, giving a near-neutral solution.
- SiCl₄ and PCl₅ are covalent. They undergo **hydrolysis** 水解 (react with water) to make acidic solutions and fumes of HCl:



Explaining the trends

These trends follow from the change in bonding and **electronegativity** 电负性. On the left, the elements are metals with low electronegativity, so their oxides and chlorides are ionic and basic (or neutral). On the right, the elements are non-metals with high electronegativity, so their oxides and chlorides are covalent and acidic. You can use a chloride's or oxide's properties (melting point, conductivity, effect on water) to suggest whether its bonding is ionic or covalent.

Periodicity of other elements

The same idea works for any group. If you know the pattern down a group and across a period, you can:

- **predict** the properties of an element from its position (for example, a Group 1 element will be a reactive metal forming a $+1$ ion).
- **deduce** the likely position and identity of an unknown element from its physical and chemical properties.